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Ảnh màn hình kết quả:

Bài 1:

```
tiem23521566@UIT:~/hadoop_lab1/bai2$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [UIT]
tiem23521566@UIT:~/hadoop_lab1/bai2$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
tiem23521566@UIT:~/hadoop_lab1/bai2$ jps
4864 NodeManager
4193 NameNode
4739 ResourceManager
4999 Jps
4487 SecondaryNameNode
4315 DataNode
tiem23521566@UIT:~/hadoop_lab1/bai2$
```

```
tiem23521566@UIT:~$ mkdir hadoop_lab1
tiem23521566@UIT:~$ cd hadoop_lab1
```

```
tiem23521566@UIT:~/hadoop_lab1$ ls
movies.txt  ratings_1.txt  ratings_2.txt  users.txt
```

```
tiem23521566@UIT:~/hadoop_lab1$ javac -classpath $(hadoop classpath) MovieRatin
gAverage.java
tiem23521566@UIT:~/hadoop_lab1$ jar -cvf movie.jar *.class
added manifest
adding: MovieRatingAverage$MapClass.class(in = 1900) (out= 765)(deflated 59%)
adding: MovieRatingAverage$ReduceClass.class(in = 3525) (out= 1531)(deflated 56
%)
adding: MovieRatingAverage.class(in = 1792) (out= 887)(deflated 50%)
tiem23521566@UIT:~/hadoop_lab1$
```

```
RESULT   Avatar (2009), Action|Adventure|Fantasy is the highest rated movie wit
h average rating 4.75
tiem23521566@UIT:~/hadoop_lab1$
```

Bài 2:

```
tiem23521566@UIT:~/hadoop_lab1$ hdfs dfs -cat /genre_output/part-r-00000
Action  Avg: 3.76, Count: 19
Adventure       Avg: 3.85, Count: 30
War     Avg: 4.00, Count: 2
Romance Avg: 3.75, Count: 8
History Avg: 4.00, Count: 6
Sci-Fi  Avg: 3.89, Count: 19
Drama   Avg: 3.81, Count: 76
Thriller        Avg: 3.88, Count: 16
Music   Avg: 3.88, Count: 4
Crime   Avg: 3.78, Count: 20
Biography       Avg: 4.14, Count: 14
Fantasy Avg: 3.95, Count: 10
Animation       Avg: 4.17, Count: 6
Family  Avg: 4.25, Count: 2
Comedy  Avg: 3.96, Count: 12
Mystery Avg: 4.00, Count: 8
Children        Avg: 4.25, Count: 2
tiem23521566@UIT:~/hadoop_lab1$
```

Bài 3:

```
tiem23521566@UIT:~/hadoop_lab1$ cat output_bai3.txt
American Beauty (1999)  Male: 3.00, Female: 4.50
Avatar (2009)   Male: 5.00, Female: 4.50
Back to the Future (1985)       Male: 4.00, Female: 4.00
Birdman (2014)  Male: 3.00, Female: 3.00
Braveheart (1995)       Male: 4.50, Female: 4.00
Coco (2017)     Male: 5.00, Female: 3.50
Dunkirk (2017)  Male: 4.00, Female: 3.00
Fight Club (1999)       Male: 5.00, Female: 3.50
Forrest Gump (1994)     Male: 2.50, Female: 5.00
Gladiator (2000)        Male: 3.50, Female: 3.00
Good Will Hunting (1997)        Male: 3.00, Female: 4.00
Gravity (2013)  Male: 4.00, Female: 4.50
Inception (2010)        Male: 3.50, Female: 4.00
Interstellar (2014)     Male: 3.50, Female: 3.50
Jurassic Park (1993)    Male: 3.50, Female: 4.50
La La Land (2016)       Male: 5.00, Female: 3.00
Mad Max: Fury Road (2015)       Male: 4.00, Female: 4.00
Manchester by the Sea (2016)    Male: 3.50, Female: 4.50
Memento (2000)  Male: 3.50, Female: 4.50
Moonlight (2016)        Male: 3.00, Female: 4.00
Parasite (2019) Male: 4.00, Female: 4.00
Pulp Fiction (1994)     Male: 4.50, Female: 3.50
Saving Private Ryan (1998)      Male: 4.00, Female: 4.00
```

```
Se7en (1995)    Male: 4.00, Female: 3.50
Slumdog Millionaire (2008)      Male: 4.50, Female: 4.00
Spotlight (2015)        Male: 4.50, Female: 4.00
Star Wars: A New Hope (1977)    Male: 4.00, Female: 3.00
The Avengers (2012)     Male: 3.50, Female: 4.00
The Big Short (2015)    Male: 5.00, Female: 4.00
The Dark Knight (2008)  Male: 4.00, Female: 3.00
The Departed (2006)     Male: 4.50, Female: 2.00
The Godfather (1972)    Male: 4.50, Female: 2.50
The Green Mile (1999)   Male: 3.00, Female: 4.00
The Imitation Game (2014)       Male: 4.50, Female: 3.00
The Lion King (1994)    Male: 4.50, Female: 3.50
The Lord of the Rings: The Fellowship of the Ring (2001)        Male: 4.50, Fem
ale: 4.00
The Lord of the Rings: The Return of the King (2003)    Male: 4.00, Female: 3.0
0
The Lord of the Rings: The Two Towers (2002)    Male: 4.50, Female: 3.00
The Matrix (1999)       Male: 3.00, Female: 4.50
The Prestige (2006)     Male: 4.00, Female: 4.00
The Revenant (2015)     Male: 3.50, Female: 3.50
The Shawshank Redemption (1994) Male: 3.00, Female: 5.00
The Silence of the Lambs (1991) Male: 3.50, Female: 3.50
The Social Network (2010)       Male: 3.50, Female: 4.00
The Usual Suspects (1995)       Male: 4.00, Female: 4.50
The Wolf of Wall Street (2013)  Male: 4.00, Female: 4.50
Titanic (1997)  Male: 4.00, Female: 3.00
Toy Story (1995)        Male: 4.50, Female: 4.00
Whiplash (2014) Male: 4.50, Female: 3.00
tiem23521566@UIT:~/hadoop_lab1$
```

Bài 4:

```
tiem23521566@UIT:~/hadoop_lab1$ hdfs dfs -cat age_output/part-r-00000
American Beauty (1999)  0-18: NA 18-35: 3.75 35-50: NA 50+: NA
Avatar (2009)   0-18: NA 18-35: 4.50 35-50: 5.00 50+: NA
Back to the Future (1985)       0-18: NA 18-35: 4.00 35-50: NA 50+: NA
Birdman (2014)  0-18: NA 18-35: 3.00 35-50: NA 50+: NA
Braveheart (1995)       0-18: NA 18-35: 4.00 35-50: 4.50 50+: NA
Coco (2017)     0-18: NA 18-35: 4.25 35-50: NA 50+: NA
Dunkirk (2017)  0-18: NA 18-35: 3.50 35-50: NA 50+: NA
Fight Club (1999)       0-18: NA 18-35: 4.25 35-50: NA 50+: NA
Forrest Gump (1994)     0-18: NA 18-35: 3.75 35-50: NA 50+: NA
Gladiator (2000)        0-18: NA 18-35: 3.00 35-50: 3.50 50+: NA
Good Will Hunting (1997)        0-18: NA 18-35: 3.50 35-50: NA 50+: NA
Gravity (2013)  0-18: NA 18-35: 4.50 35-50: 4.00 50+: NA
Inception (2010)        0-18: NA 18-35: 3.75 35-50: NA 50+: NA
Interstellar (2014)     0-18: NA 18-35: 3.50 35-50: 3.50 50+: NA
Jurassic Park (1993)    0-18: NA 18-35: 4.00 35-50: NA 50+: NA
La La Land (2016)       0-18: NA 18-35: 3.00 35-50: 5.00 50+: NA
Mad Max: Fury Road (2015)       0-18: NA 18-35: 4.00 35-50: NA 50+: NA
Manchester by the Sea (2016)    0-18: NA 18-35: 4.50 35-50: 3.50 50+: NA
Memento (2000)  0-18: NA 18-35: 4.50 35-50: 3.50 50+: NA
Moonlight (2016)        0-18: NA 18-35: 4.00 35-50: 3.00 50+: NA
Parasite (2019) 0-18: NA 18-35: 4.00 35-50: 4.00 50+: NA
Pulp Fiction (1994)     0-18: NA 18-35: 4.00 35-50: NA 50+: NA
Saving Private Ryan (1998)      0-18: NA 18-35: 4.00 35-50: 4.00 50+: NA
Schindler's List (1993) 0-18: NA 18-35: 4.50 35-50: 4.00 50+: NA
Se7en (1995)    0-18: NA 18-35: 3.75 35-50: NA 50+: NA
Slumdog Millionaire (2008)      0-18: NA 18-35: 4.00 35-50: 4.50 50+: NA
Spotlight (2015)        0-18: NA 18-35: 4.00 35-50: 4.50 50+: NA
```

```
Spotlight (2015)        0-18: NA 18-35: 4.00 35-50: 4.50 50+: NA
Star Wars: A New Hope (1977)    0-18: NA 18-35: 3.00 35-50: 4.00 50+: NA
The Avengers (2012)     0-18: NA 18-35: 3.75 35-50: NA 50+: NA
The Big Short (2015)    0-18: NA 18-35: 4.50 35-50: NA 50+: NA
The Dark Knight (2008)  0-18: NA 18-35: 3.00 35-50: 4.00 50+: NA
The Departed (2006)     0-18: NA 18-35: 2.00 35-50: 4.50 50+: NA
The Godfather (1972)    0-18: NA 18-35: 3.50 35-50: NA 50+: NA
The Green Mile (1999)   0-18: NA 18-35: 4.00 35-50: 3.00 50+: NA
The Imitation Game (2014)       0-18: NA 18-35: 3.00 35-50: 4.50 50+: NA
The Lion King (1994)    0-18: NA 18-35: 4.00 35-50: NA 50+: NA
The Lord of the Rings: The Fellowship of the Ring (2001)        0-18: NA 18-35:
 4.00 35-50: 4.50 50+: NA
The Lord of the Rings: The Return of the King (2003)    0-18: NA 18-35: 3.50 35
-50: NA 50+: NA
The Lord of the Rings: The Two Towers (2002)    0-18: NA 18-35: 3.00 35-50: 4.5
0 50+: NA
The Matrix (1999)       0-18: NA 18-35: 3.75 35-50: NA 50+: NA
The Prestige (2006)     0-18: NA 18-35: 4.00 35-50: 4.00 50+: NA
The Revenant (2015)     0-18: NA 18-35: 3.50 35-50: NA 50+: NA
The Shawshank Redemption (1994) 0-18: NA 18-35: 4.00 35-50: NA 50+: NA
The Silence of the Lambs (1991) 0-18: NA 18-35: 3.50 35-50: 3.50 50+: NA
The Social Network (2010)       0-18: NA 18-35: 4.00 35-50: 3.50 50+: NA
The Usual Suspects (1995)       0-18: NA 18-35: 4.25 35-50: NA 50+: NA
The Wolf of Wall Street (2013)  0-18: NA 18-35: 4.50 35-50: 4.00 50+: NA
Titanic (1997)  0-18: NA 18-35: 3.00 35-50: 4.00 50+: NA
Toy Story (1995)        0-18: NA 18-35: 4.00 35-50: 4.50 50+: NA
Whiplash (2014) 0-18: NA 18-35: 3.00 35-50: 4.50 50+: NA
tiem23521566@UIT:~/hadoop_lab1$
```